# Nemotron

**Planned integration**

A bounded investigation agent that reads incident evidence, explains uncertainty, and recommends the next step. The backend remains responsible for approval and service actions.

## Integration boundary and evidence

The deployed backend currently opens incidents and creates inspection requests with deterministic rules. The owner dashboard works with those requests today. A live model endpoint, tool runner, and persisted investigations remain to be implemented.

| Evidence | Interpretation |
|---|---|
| **Office light** | Read expected lighting state, reported LED state, threshold, fresh samples, and duration. Shadows and surrounding light can change the signal. The production threshold is pending bench calibration. |
| **Data-center flame/IR** | Distinguish active and brief activations. Consider freshness and any inspection notes. An activation can justify inspection without identifying its cause. |
| **Temperature and fan** | Report the room estimate and its source or preserve the raw count. Temperature is supporting context. Fan state is a controller request, with no speed or motion measurement. |
| **Repair cases** | Search for comparable equipment, issue category, verified outcome, and provenance. Cite relevant case IDs or say when none match. |

## First model smoke test

Choose the exact deployed Nemotron model and endpoint, confirm it returns structured tool calls, and execute one read-only request for a known incident. Feed the result back and verify that the final recommendation reflects the returned evidence.

NVIDIA NIM tool calling documentation

# Proposed tools and run state

## NEMOTRON

| Tool surface | Contract |
| --- | --- |
| **get_incident** | Load the scoped incident, building, source, recorded evidence and current approval state. |
| **get_sensor_history** | Return bounded recent samples with units, capture times, freshness and estimated-temperature status. |
| **search_verified_repairs** | Return relevant cases with equipment, issue kind, finding, verification method and provenance. |
| **request_owner_clarification** | Persist a concise question when the evidence is ambiguous. Resume from the verified reply. |
| **propose_inspection** | Return evidence, proposed work, uncertainty, and any relevant repair IDs for backend validation. |
| **get_approval_status** | Read the current backend decision when an investigation resumes. |

Use a bounded worker that validates tool names, arguments, incident scope, and result sizes. Persist phase, tool trace, observations, latest owner summary and waiting state. Cap turns and time. On restart, reload current incident and approval state before resuming.

The backend validates any proposed service request and owns versioning, expiry and dispatch authority. Keep external sends behind Photon and the durable job queue. The agent can recommend monitoring, ask a question, propose inspection or escalate for human review.

# Scenarios and exit checks

**NEMOTRON**

## Office lighting

Compare recent readings with the measured normal range when the lights are expected and reported on. If evidence is stale or the sensor may be obstructed, request clarification or more readings. If the dip is sustained, recommend inspecting the lighting circuit. Explain that a specific failed component remains unconfirmed.

## Data-center IR

Describe whether the activation is present, repeated or brief. Check recent context and comparable verified cases. Propose a manual inspection when appropriate. A clear follow-up input can be recorded as evidence, with human verification required for closure.

## MVP acceptance

**01** A real model run makes an evidence-dependent branch and produces a traceable, scoped tool call.

**02** Test fresh and stale telemetry, relevant and unrelated repair cases, a covered light sensor, and an IR pulse. Include a labeled simulator fixture for repeatable rehearsals.

**03** Proposals can wait for an owner decision, survive a worker restart and resume against the current version. A failed run leaves the existing dashboard workflow usable.

**04** A declined request leaves the incident open and sends no technician job. Approved work advances through the same backend gate that serves the dashboard. Record verified outcomes for later retrieval.